ROSHAN RAJ MAHATO

+91-9204775882 — roshanrajmahato1234@gmail.com — linkedin.com/in/roshanrajmahto/

INTERNSHIP

Medical Chatbot Developer Intern
YBI Foundation (Remote) *Jun – Jul 2024*

- Developed a Medical Chatbot leveraging the **Llama-2-7B-Chat** model for accurate, context-aware medical information retrieval.
- Utilized **LangChain Framework** to streamline generative AI workflows, enabling seamless integration of the Llama-2 model and Pinecone database for efficient medical query handling.
- Integrated **Pinecone Vector Database** for efficient storage and retrieval of embeddings, enhancing response time and accuracy.
- Built a user-friendly web interface using **Flask, HTML, and CSS** with a modular backend for maintainable and scalable code structure.

SOFT SKILLS

- Fast Learner
- Time Management
- Communication Skills
- Problem Solving

TECHNICAL SKILLS

Programming Languages:

- **Python** (OOP, Data Structures, NumPy, Pandas, Matplotlib)
- **C++** (Basics, OOP, Data Structures)

Machine Learning:

- Logistic Regression, SVM, Random Forest, XGBoost & other algorithms
- Scikit-learn, Hyperparameter Tuning, Feature Engineering

Deep Learning:

- TensorFlow, PyTorch, CNNs, RNNs, Transformers

Generative AI:

- LangChain, RAG, LLMs, OpenAI API

Data Science:

- NumPy, Pandas, Matplotlib, Statistical Analysis

AI / MLOps:

- MLflow, Docker, AWS (EC2, ECR), GCP (Cloud Functions, Cloud Tasks, Buckets), Git, CI/CD

EDUCATION

Ramgarh Engineering College *Jun 2021 – Jul 2025*
Bachelor of Engineering in Computer Science & Engineering
CGPA: 7.0

TOOLS

- **IDE & Code Editors:** Proficient in Visual Studio Code and PyCharm for efficient development and debugging.
- **MLOps & Cloud:** Experienced with MLflow for experiment tracking; Docker for containerisation; AWS (EC2, ECR) and GCP (Cloud Functions, Cloud Tasks) for deployment.
- **Version Control:** Experienced in using Git and GitHub for collaborative development, code management, and CI/CD pipelines.

PROJECTS

Student Performance Prediction *Dec 2023*
Python, Scikit-learn, Machine Learning

- Achieved a **96% accuracy** rate in forecasting student academic performance by developing and deploying a machine learning model.
- Managed data integrity by handling missing values and encoding categorical variables, enhancing data quality by 33%.
- Conducted experiments with both classification and regression algorithms to identify the most suitable approach.

Credit Card Fraud Detection *Oct 2023*
Python, Logistic Regression, Scikit-learn

- Developed and fine-tuned a logistic regression-based ML model achieving an **87% accuracy** rate.
- Analyzed transaction patterns to identify anomalies and distinguish between genuine and fraudulent activities.
- Optimized detection efficiency, contributing to the potential prevention of financial losses in real-world scenarios.

Pneumonia Detection and Classification *Sept 2022*
Deep Learning, TensorFlow, Keras, VGG16, Transfer Learning

- Designed a Deep Learning model using **VGG16 with Transfer Learning** to classify chest X-ray images as Pneumonia or Normal, achieving **91% accuracy**.
- Leveraged Keras and TensorFlow to build and train a Convolutional Neural Network, enhancing medical imaging diagnostics through automated detection.